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DATA DESCRIPTOR

GSOCS-LULCC: A Global dataset of Soil Organic Carbon Stock dynamics following Land Use and Land Cover Change

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The direction and magnitude of soil organic carbon stock (SOCS) changes following land use and land cover change (LULCC) remain highly uncertain, primarily due to a lack of relevant, comprehensive global soil data. Although substantial efforts have established SOCS databases at regional, national, and sub-continental scales, an open-access global database incorporating explicit environmental drivers is urgently needed to better understand the impacts of LULCC on SOCS dynamics. Here, we present the Global Soil Organic Carbon Stock dataset after Land Use and Land Cover Change (GSOCS-LULCC). Compiled from 912 peer-reviewed articles up to December 2025, this dataset spans five major land uses (cropland, grassland, forest, plantation and savanna) yielding 8,748 harmonized records from 1,967 sites across various sampling depths. GSOCS-LULCC serves as a vital resource for evaluating the global effects of LULCC on SOCS and identifying the primary environmental factors driving carbon stock variations across different ecosystem transitions.

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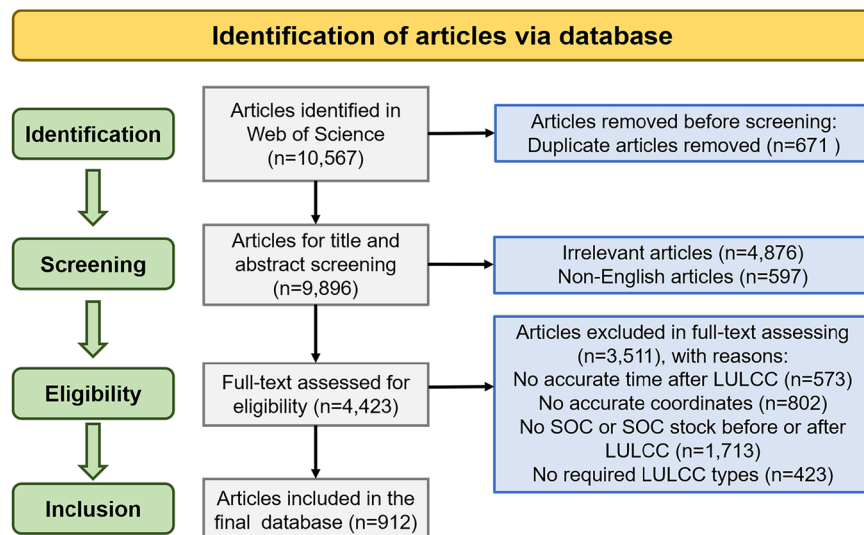


Fig. 1 The PRISMA flow diagram for articles identification and selection.

Background & Summary

Soil organic carbon (SOC) is essential for regulating the physical, chemical, and biological properties of soil, thereby directly driving soil functions. As the largest carbon (C) reservoir in terrestrial ecosystems (~1,550 Pg), SOC stores nearly twice the amount of carbon found in the atmosphere (~760 Pg) and roughly 2.8 times the amount stored in global vegetation (~560 Pg)¹. Because the terrestrial SOC pool is substantially larger than the atmospheric carbon pool, even minor relative fluctuations in these immense SOC reservoirs translate to massive absolute changes in atmospheric carbon concentrations, thereby profoundly altering global C cycling and significantly influencing climate change trajectories².

Ecosystem transitions driven by LULCC disrupt the historical equilibrium between carbon inputs and outputs, leading to shifts in SOCS until a new steady state is achieved^{3,4}. The magnitude and direction of these changes—whether triggering carbon accrual or carbon losses—are governed by complex interactions among climate conditions, soil physicochemical properties, vegetation types, and land management history^{5–8}. Numerous studies have highlighted the vulnerability of SOCS to LULCC^{3,9,10}. Moreover, recent global assessments, such as the Global Carbon Budget 2025, has underscored the persistent uncertainty in estimating LULCC-induced carbon emissions¹¹. Achieving modern global climate targets, such as those outlined in the Paris Agreement and the 4 per 1000 initiative, relies heavily on mitigating these uncertainties and optimizing land management for verifiable carbon sequestration.

Despite the wealth of existing localized research on these dynamics, a systematic, open-access global database linking SOCS changes to explicit environmental drivers has remained largely unavailable¹². This critical absence limits the scientific community's ability to assess long-term geospatial trends and refine global carbon cycle models. To address this gap, we developed the GSOCs-LULCC by systematically synthesizing data from high-quality, peer-reviewed literature. This robust dataset establishes a comprehensive foundation for understanding and predicting SOCS responses to global LULCC, ultimately supporting more accurate assessments of terrestrial carbon fluxes and informing evidence-based climate change mitigation strategies.

Methods

Literature Search and Selection. We compiled literature on SOCS changes following LULCC from English-language peer-reviewed journals indexed in the Web of Science up to December 2025. The literature search was conducted with the following keyword query: “TITLE: (land use change OR land use dynamic OR land use conversion OR land cover change OR land cover dynamic OR land cover conversion OR deforestation OR afforestation OR reforestation OR restoration OR plantation) AND TOPIC: (soil carbon OR soil organic carbon OR soil organic matter)”¹³. Article selection strictly adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines across four phases: identification, screening, eligibility, and inclusion¹⁴ (Fig. 1).

From an initial pool of 10,567 articles, 671 duplicates were removed prior to screening. Subsequent title and abstract screening excluded 4,876 irrelevant studies and 597 non-English articles. We then conducted a rigorous full-text eligibility assessment on the remaining 4,423 articles. During this phase, studies were excluded if they lacked post-LULCC temporal data (n = 573), omitted precise geographic coordinates (n = 802), missed essential pre- or post-LULCC SOC/SOCS data (n = 1,713), or did not evaluate the required land-use categories (n = 423). Ultimately, 912 articles met all inclusion criteria, yielding a final comprehensive database of 8,748 records across 1,967 global sites.

Data Extraction and Harmonization. From the selected 912 articles, data extraction was stratified into two categories to distinguish between strictly empirical measurements and variables subject to spatial gap-filling.

Land use	Definition
Cropland	Site described as annual or perennial crop cultivation, but excluding temporarily fallow areas
Grassland	Site described as natural or semi-natural herbaceous vegetation (natural grassland), or managed pastures, grazing lands, or hayfields (managed grassland)
Forest	Site described as primary, old-growth, native, or unmanaged secondary forests
Plantation	Site description indicated active silviculture, timber harvesting, or commercial tree cropping
Savanna	Site described as grass-dominated ecosystems with scattered trees, typically in tropical or subtropical regions with seasonal rainfall

Table 1. Land-use specification.

Mandatory extracted data: The following minimum required data were extracted exclusively from the primary literature: (1) Basic article information (publication year, authors, title, journal); (2) Precise geographic coordinates (latitude and longitude) and country; (3) Soil sampling design (paired-site, repeated sampling, or chronosequence) and time since LULCC; (4) Sampling depth; and (5) Utilization of mass correction via the equivalent soil mass (ESM) approach to adjust SOC stock change estimates¹⁰.

Ancillary and supplemented data: The following environmental and site-specific variables were extracted from the manuscripts if reported; otherwise, they were gap-filled using external geospatial sources: (1) Mean and standard deviation (or standard error) of pre- and post-transition SOC or SOCS; (2) Topographic relief (elevation); (3) Climate and vegetation metrics (MAT, MAP, PET, and GPP); and (4) Soil physicochemical characteristics (WRB soil type, pH, particle size distribution (clay, silt, sand), and bulk density).

Data were systematically extracted from article texts, tables, figures, and supplementary materials. When relevant information was restricted to graphical formats, we utilized the GetData Graph Digitizer to accurately extract the underlying data points. The database focuses specifically on five primary land-use types: cropland, grassland, forest, plantation, and savanna. Explicit definitions and inclusion criteria for each land-use category are detailed in Table 1.

All soil data represent mineral soil, excluding the litter (organic) layer, which is known to alter the directionality of SOC dynamics¹⁵. It is critical to note that no mean pre-LUC or post-LUC SOC concentrations or SOCS values were gap-filled. All baseline and post-transition target values are strictly empirical measurements directly extracted from the literature. Gap-filling was strictly limited to ancillary variables (climate, vegetation, elevation, soil type, bulk density, pH, particle size distribution, and standard deviations of SOC or SOCS) extracted from external global databases using the precise latitude and longitude reported for each study plot. Climate metrics (MAP and MAT) were extracted from WorldClim Version 2.0 (1 km resolution)¹⁶, while PET was sourced from the Global Aridity Index and Potential Evapotranspiration Climate Database v3 (1 km resolution)¹⁷. Elevation data were derived from the SRTM 90 m Digital Elevation Database v4.1¹⁸, and GPP was extracted from the VODCA2GPP product¹⁹. Soil properties (soil type, pH, and particle size distribution) were retrieved from SoilGrids 2.0 at a 250 m resolution across six standard GlobalSoilMap depth intervals (0–5, 5–15, 15–30, 30–60, 60–100, and 100–200 cm)²⁰. These continuous geospatial values were harmonized to corresponding profile depth intervals utilizing numerical integration^{21–23}. For sampled sites reporting SOC content but lacking bulk density measurements, we estimated bulk density using a pedotransfer function developed by Tao *et al.*²⁴, derived from WoSIS data encompassing 78,913 soil layers²⁵.

$$BD = 0.32 + 1.30 \times e^{0.0089 \times SOM} \quad (1)$$

where BD is the bulk density (g cm^{-3}), SOM is the soil organic matter content (%) calculated as $\text{SOC} \times 1.724$.

SOC content (g kg^{-1}) was subsequently converted to SOCS (t ha^{-1}) using the following equation²⁶:

$$\text{SOCS} = \text{SOC} \times BD \times \text{Thickness} \times 0.1 \quad (2)$$

where Thickness is the thickness of soil layer (cm), and 0.1 is the conversion factor for unit.

For datasets omitting standard deviation or standard error for SOC or SOCS, we conservatively assigned a standard deviation equal to one-tenth of the respective mean, in accordance with established methodological precedents^{27,28}.

Preliminary Predictive Modeling of SOCS Dynamics. To quantitatively evaluate SOCS responses to LULCC under varying environmental conditions, we evaluated three distinct machine learning algorithms: Random Forest (RF), Gradient Boosting Machine (GBM), and Cubist. The Cubist algorithm consistently exhibited superior predictive performance and was thus selected as the primary modeling framework. As a robust non-parametric method, Cubist fits multivariate linear models at the terminal leaves of a decision tree and utilizes committee models alongside instance-based nearest-neighbor corrections to optimize both predictive accuracy and interpretability. Such rule-based ensemble methods are increasingly recognized in digital soil mapping for their ability to untangle complex, non-linear soil-environment relationships^{29,30}.

For model development, the dataset was stratified by LULCC transition type. Within each transition category, data were randomly partitioned into a training set (70%) and an independent validation set (30%). Models incorporated 12 environmental and site-specific predictors—MAT, MAP, PET, elevation, GPP, clay/silt/sand fractions, pH, time since LULCC, initial SOCS, and mean sampling depth—to predict the target variable: absolute SOCS change (post-LULCC minus pre-LULCC).

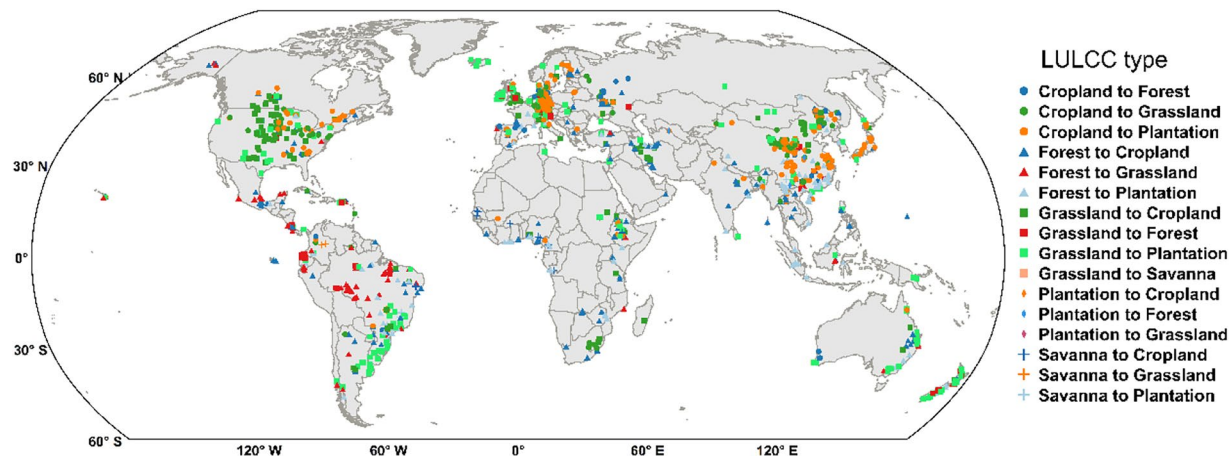


Fig. 2 Spatial locations of study sites after five major types of LULCC: (a) cropland to others (circle); (b) forest to others (triangle); (c) grassland to others (square); (4) plantation to others (diamond); (5) savanna to others (cross).

All Cubist models were implemented using the cubist package in R³¹. Model performance on the validation set was assessed coefficient of determination (R^2), root mean square error (RMSE), and bias.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (4)$$

$$Bias = \hat{y}_i - y_i \quad (5)$$

where y_i and $\hat{y}_i$ are observation and prediction for sample i , $\bar{y}$ is the mean of observations, and n is the number of samples.

Data Records

The dataset contains 8,748 harmonized records sourced from 912 peer-reviewed articles. The database is structured across 61 columns, capturing comprehensive information regarding publication metadata, spatial coordinates, climatic indices, vegetation metrics, topographic relief, and detailed soil properties. To ensure data provenance transparency, dedicated indicator columns specify whether a given variable was directly extracted or gap-filled using external sources. Detailed definitions for all variables are meticulously outlined in the accompanying README file. The complete dataset is publicly hosted and accessible on Zenodo (<https://doi.org/10.5281/zenodo.20774640>)³².

Technical Validation

The GSOCS-LULCC database captures a robust spatial distribution across 16 primary LULCC transition types (Fig. 2): (1) Cropland to: forest (456 records), grassland (1,194 records), plantation (1,803 records); Forest to: cropland (1,011 records), grassland (759 records), plantation (1,041 records); Grassland to: cropland (719 records), forest (339 records), plantation (1,159 records), savanna (8 records); Plantation to: cropland (57 records), forest (16 records), grassland (19 records); Savanna to: cropland (61 records), grassland (19 records), plantation (87 records).

While pre-transition forest and grassland sites demonstrate a broad global distribution, cropland transitions are predominantly concentrated in the United States, Europe, and China. Conversely, savanna transitions are distinctly localized to Africa and South America. Ecologically, the dataset represents nearly all Whittaker biomes—with notable measurement densities in temperate seasonal forests and tropical savannas—and comprehensively covers the USDA soil texture triangle, particularly within the clay loam and loam classifications (Fig. 3).

Descriptive statistics indicate that over 80% of the sampled profiles maintained SOCS values below 50 t ha⁻¹ both prior to and following LULCC (Fig. 4). Overall, the mean SOCS exhibited a slight post-transition decrease, shifting from 33.01 t ha⁻¹ to 31.53 t ha⁻¹. Evaluating specific transition dynamics, conversions to forest consistently enhanced SOCS, increasing carbon stocks when initiated from cropland (+4.74 t ha⁻¹, +21.95%), grassland (+2.39 t ha⁻¹, +9.71%), and plantation (+1.95 t ha⁻¹, +7.88%).

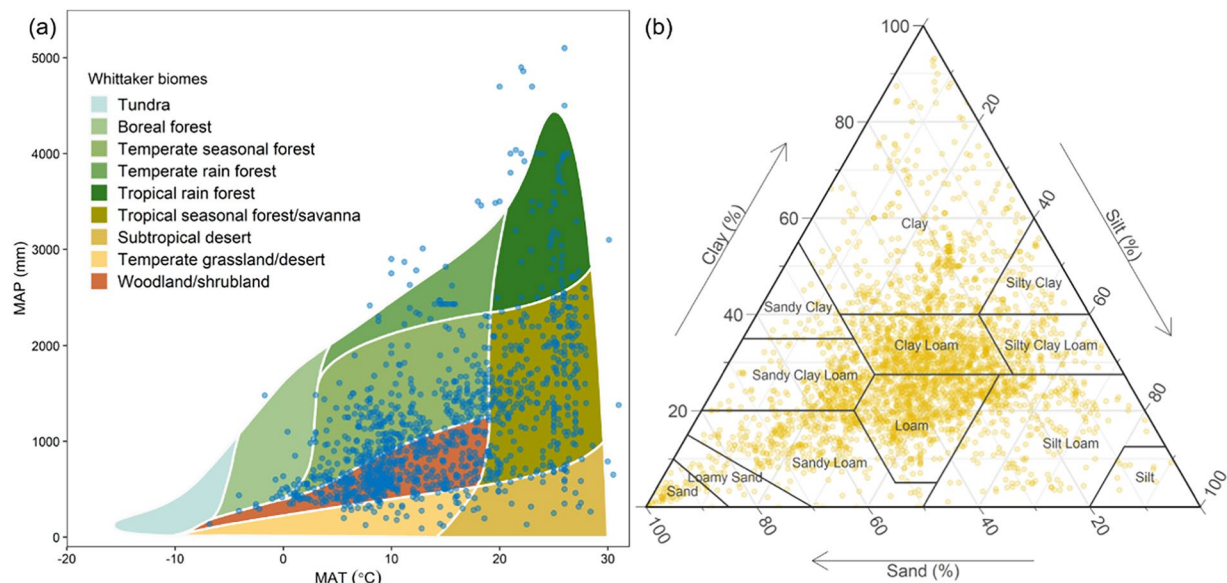


Fig. 3 The coverages of GSOCS-LULCC on Whittaker biomes (a) and USDA soil texture triangle (b).

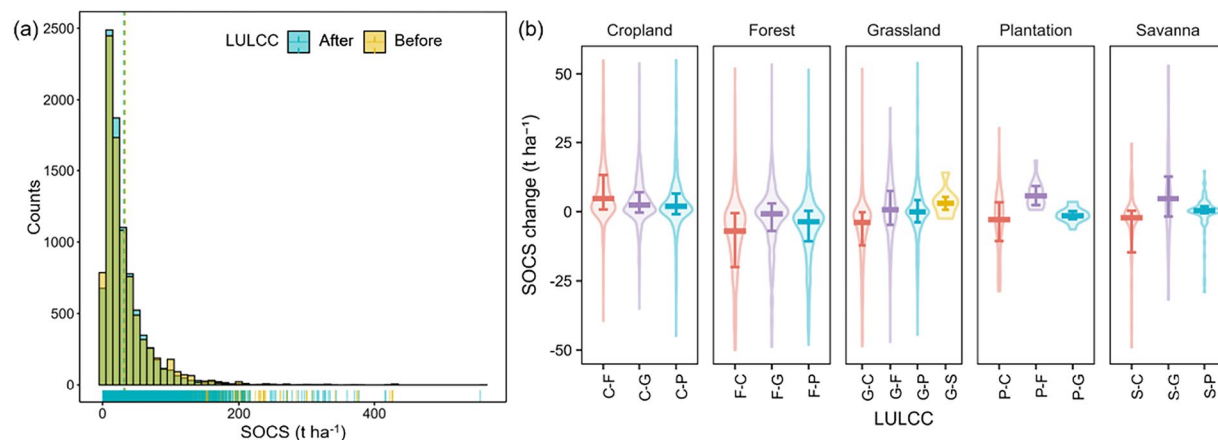


Fig. 4 Histograms of SOCS (in t ha^{-1}) before and after LULCC recorded in the GSOCS-LULCC (a) and its change for each LULCC (b). The vertical dash lines (in a) show the mean of SOCS before and after LULCC, and the red horizontal dash line (in b) is at the 0. The abbreviations (in b) are listed below: C, cropland; F, forest; G, grassland; P, plantation; S, savanna.

Conversely, anthropogenic conversions generally depleted SOCS. Shifting forest to cropland yielded a significant decrease (-6.99 t ha^{-1} , -13.79%), as did forest to plantation (-3.65 t ha^{-1} , -8.45%), while forest to grassland resulted in a negligible change (-0.75 t ha^{-1} , -2.04%). Substantial declines were also observed when grassland was converted to cropland (-3.90 t ha^{-1} , -10.31%). Interestingly, converting grassland to savanna resulted in a substantial increase ($+3.04 \text{ t ha}^{-1}$, $+34.19\%$), whereas conversions to forest ($+0.71 \text{ t ha}^{-1}$, $+2.01\%$) and plantation (-0.05 t ha^{-1} , -0.17%) exhibited minimal change. For plantation conversions, SOCS decreased when shifted to cropland (-2.87 t ha^{-1} , -10.88%) and grassland (-1.47 t ha^{-1} , -12.02%), but significantly increased when converted to forest ($+5.72 \text{ t ha}^{-1}$, $+36.58\%$). Savanna conversions showed increased SOCS when transitioning to grassland ($+4.70 \text{ t ha}^{-1}$, $+7.63\%$), slight decreases when converting to cropland (-2.18 t ha^{-1} , -5.54%), and no significant change towards plantations ($+0.39 \text{ t ha}^{-1}$, $+2.76\%$).

Temporal and depth analyses reveal critical gaps in current literature (Fig. 5). While the dataset spans a broad temporal range (from <5 years to >100 years post-LULCC), 72% of the samples capture relatively short-term dynamics (<30 years). This highlights inherent uncertainties when extrapolating long-term SOCS trajectories. Furthermore, over two-thirds of the measurements are restricted to the topsoil ($\leq 30 \text{ cm}$), with deep-soil measurements ($>100 \text{ cm}$) accounting for less than 1% of the dataset. This emphasizes an urgent need for deeper profile sampling in future research campaigns to fully comprehend holistic SOCS dynamics.

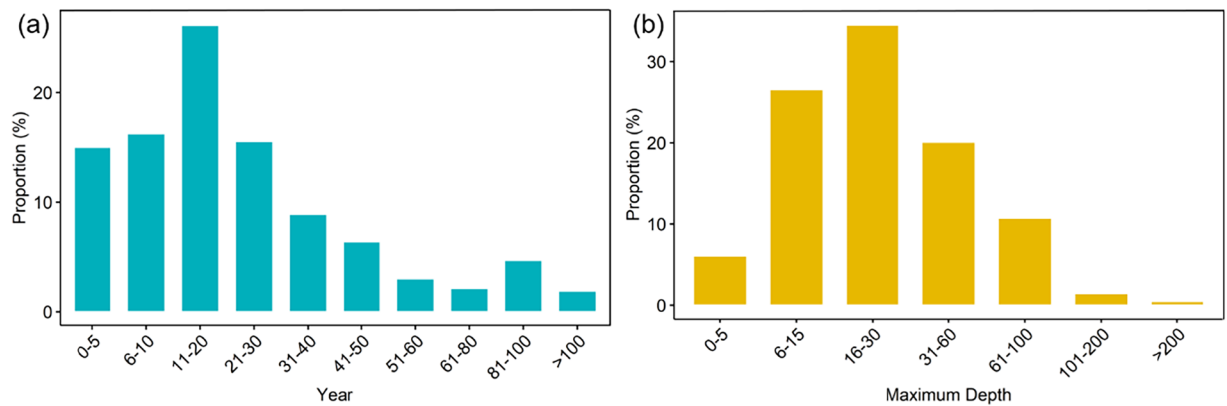


Fig. 5 The time after LULCC (a) and the maximum soil depth (in cm) of interest (b) in the GSOCS-LULCC. The year is split into the intervals of 0-5, 6-10, 11-20, 21-30, 31-40, 41-50, 51-60, 61-80, 81-100 and >100. The maximum depth is classified into 0-5, 6-15, 16-30, 31-60, 61-100, 101-200 and >200 based on GlobalSoilMap specifications.

Key variables	Proportion (%)	Key variables	Proportion (%)
Soil type (WRB)	100	Clay	64.4
PET	91.9	Elevation	46.9
Silt	69.6	Bulk density start	45.2
Sand	69.2	Bulk density end	47.1
pH	67.9	MAT	20.2

Table 2. The top 10 variables required spatial gap-filling (in proportion).

To transparently quantify the reporting gaps in primary literature and highlight the provenance of our predictors, Table 2 details the proportion of data directly extracted from manuscripts versus data derived from spatial gap-filling. Notably, variables such as soil type (in WRB), PET, soil particle distribution (i.e., clay, silt, and sand), pH, bulk density, frequently required gap-filling, underscoring a critical need for more comprehensive reporting in future LULCC field studies.

While this database provides a comprehensive global overview, users must account for inherent uncertainties related to historical land-use legacies. Primary literature rarely reports the pre-LUC equilibrium status of study sites. Consequently, sites that had not reached SOC equilibrium prior to the documented transition may exhibit SOC dynamics muddled by historical effects. In addition, due to aggregation necessities, categories such as grassland encompass both natural/unmanaged grasslands and intensively managed European pastures. Similarly, the legacy of intensively managed or replanted forests (e.g., non-native tree cropping) introduces significant variance in baseline carbon inputs and subsequent SOC changes. Furthermore, regional discrepancies in ecosystem definitions present a challenge. For example, many European forests considered natural within their regional context have undergone extensive historical management or replanting over the last century. Because such historical legacies are often omitted from contemporary site descriptions, it is likely that some sites classified as forest carry undocumented management histories, potentially altering their baseline SOC dynamics compared to undisturbed primary ecosystems.

Predictive validation of the Cubist models across the 16 LULCC types demonstrated robust utility (Fig. 6). Model performance varied, with R^2 ranged from 0 to 0.85 (mean = 0.44), RMSE from 2.28 to 17.52 (mean = 10.38), and bias from -3.90 to 2.54 (mean = -0.31). Peak predictive ability was observed for Savanna-to-Cropland transitions ($R^2 = 0.85$, RMSE = 8.90, bias = 1.81), followed by Forest-to-Cropland ($R^2 = 0.78$, RMSE = 11.97, bias = 0.67) and Savanna-to-Plantation ($R^2 = 0.76$, RMSE = 3.53, bias = 0.59). In contrast, poor performance was observed for Savanna-to-Grassland, Forest-to-Plantation and Grassland-to-Plantation ($R^2 < 0.2$), directly highlighting data-poor transitions that require targeted future sampling or alternative modeling architectures. The remaining models showed satisfactory performance with $R^2 > 0.25$, suggesting that the Cubist algorithm can reliably capture the direction and magnitude of SOCS changes for the majority of LULCC types. Ultimately, these results validate the potential of machine learning approaches for explicit geospatial predictions of LULCC-induced SOCS dynamics.

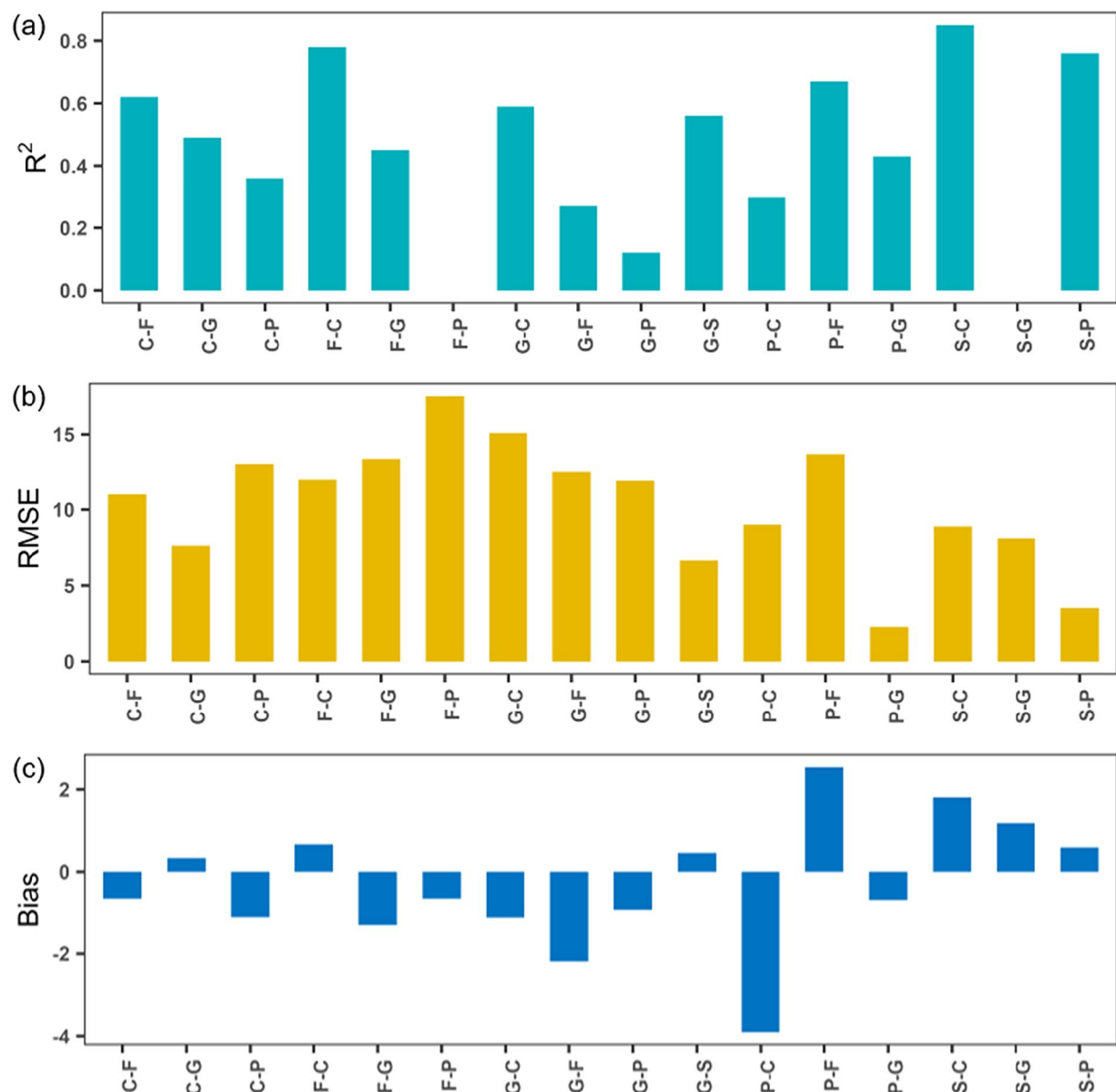


Fig. 6 Performance of Cubist prediction models for each LULCC type (a) R^2 (b) RMSE (c) Bias. When R^2 is less than 0, it is treated as 0 during plotting.

Data availability

The Global Soil Organic Carbon Stock dataset after Land Use and Land Cover Change (GSOCS-LULCC) used in this study is publicly available at Zenodo (<https://doi.org/10.5281/zenodo.20774640>). The dataset is provided in CSV format. Additional details on the dataset contents and variables are described in the Data Records section.

Code availability

The source code for model training, and validation is publicly available at Zenodo (<https://doi.org/10.5281/zenodo.20774640>).

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Author contributions

QS, DJ, LD, YH, NW, JX and SC compiled the data. QS and SC performed the analysis and drafted the manuscript. DJ, DA, ZC, LD, YD, YH, BH, YH, WJ, SL, ZL, YM., ACR-d-F, CS, YS, HT, NW, XW, YW, ZW, DX, JX, SY, XZ, YZ, PZ, and ZS validated the data, results, and revised the manuscript. SC and ZS acquired of the financial support. SC supervised this work.

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Competing interests

The authors declare no competing interests.

Additional information

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